LOTQ-0008 / FUNCTIONS / INTERMEDIATE / FIELD GUIDE

Functions Expanded Field Guide

Function families, formulas, graph behavior, research uses, and common traps.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0008
CATEGORY	Functions
DIFFICULTY	Intermediate
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

How to use this manual

Read as a field guide, not as a textbook.

1. USE THE TOP FORMULA FIRST

Start each card by reading the one-sentence meaning, then the formula block. The formula tells you the skeleton; the rest of the card tells you where it breaks, how it behaves, and how it fits in a quant workflow.

2. CONNECT SHAPE TO RESEARCH USE

Do not memorize functions as isolated school objects. Tie every graph shape to a trading task: transform a feature, model uncertainty, encode a cycle, gate a signal, smooth a curve, or control a position.

3. TRACK DOMAIN, RANGE, AND FAILURE MODES

Most bugs come from invalid inputs, boundary behavior, overflow, discontinuities, rounding, or software conventions. Every transform should have unit tests and metadata.

4. TREAT FUNCTIONS AS SYSTEM COMPONENTS

In a research system, a function is a reproducible transform node. Store parameters, fitting window, source data, output shape, and validation checks so the same signal can be rebuilt later.

5. REVIEW BY FAMILIES

Algebraic functions give curve vocabulary. Piecewise functions give rules. Exp/log functions give compounding and scaling. Trig/waves give cycles. Probability functions give uncertainty. Special functions support numerical modeling.

Function taxonomy map

How the expanded list is organized into usable research families.

1. Core Algebraic Families

20 concepts: Linear, Constant, Identity, Quadratic, Cubic, Quartic...

2. Piecewise, Discrete, and Threshold Functions

17 concepts: Piecewise, Step, Floor, Ceiling, Greatest integer, Nearest integer...

3. Exponential, Logarithmic, and Saturating Functions

10 concepts: Exponential, Natural exponential, Exponential growth, Exponential decay, Logarithmic, Natural logarithm...

4. Trigonometric and Inverse Trigonometric Functions

12 concepts: Sine, Cosine, Tangent, Cotangent, Secant, Cosecant...

5. Hyperbolic Functions

9 concepts: Hyperbolic sine, Hyperbolic cosine, Hyperbolic tangent, Hyperbolic cotangent, Hyperbolic secant, Hyperbolic cosecant...

6. Periodic, Oscillatory, and Wave Functions

7 concepts: Periodic, Oscillatory, Sinusoidal, Wave, Square wave, Triangle wave...

7. Function Properties and Mapping Behavior

14 concepts: Even, Odd, Increasing, Decreasing, Monotonic, Strictly increasing...

8. Continuity, Discontinuity, and Smoothness

8 concepts: Continuous, Discontinuous, Removable discontinuity, Jump discontinuity, Infinite discontinuity, Smooth...

9. Composition, Representations, Sequences, and Series

12 concepts: Composite, Inverse, Implicit, Explicit, Parametric, Polar...

10. Probability and Distribution Functions

7 concepts: Probability density, Cumulative distribution, Normal distribution, Uniform distribution, Binomial distribution, Poisson distribution...

11. Special Functions, Approximation, and ML Activations

23 concepts: Gamma, Beta, Zeta, Factorial, Double factorial, Gamma extension...

Notation and implementation baseline

Use this page to standardize formulas and code thinking.

Symbol	Meaning in this manual
`x`	input variable, feature value, or independent variable
`f(x)`	output of the function at input x
`a, b, c`	generic coefficients or bounds depending on context
`n, k`	integer index, degree, count, or sequence step
`t`	time index or continuous time variable
`P_t`	price at time t
`r_t`	return at time t; often simple or log return
`mu`	mean or location parameter
`sigma`	standard deviation or scale parameter
`lambda`	arrival intensity or exponential decay rate
`tau`	threshold or cutoff value
`theta`	angle, phase parameter, or model parameter
`domain`	valid inputs
`range`	actual outputs produced
`codomain`	declared target output set

IMPLEMENTATION RULE

Every function in a research system should expose: name, formula, parameters, domain check, vectorized implementation, test inputs, failure behavior, and whether it was fit on training data only.

Table of contents

Section-level navigation plus concept counts and starting pages.

MANUAL STRUCTURE

This PDF contains 139 concept cards. Each card follows the same vertical path: meaning, formula, behavior, quant use, system fit, related terms, traps, and quick recall.

SECTION 01

Core Algebraic Families

Algebraic families are the base vocabulary for curve shape, regression basis functions, cost curves, constraints, and feature transforms.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Linear

DIFFICULTY: INTERMEDIATE

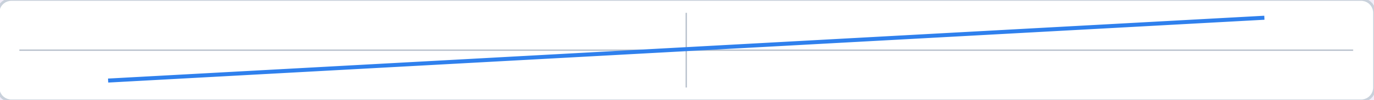
MEANING AND MEMORY

Meaning: A first-degree rule where output changes by a constant amount for each unit change in input.
Memory jogger: Straight-line change.

FORMULA AND BEHAVIOR

```
f(x) = m*x + b
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: next_return_estimate = b0 + b1*momentum_score. The slope b1 tells how much the forecast moves per score unit.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not call every straight-looking relationship linear if the intercept, scale, or residual structure is ignored.
Related terms: Identity, affine model, slope, intercept, regression

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Linear without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

CORE ALGEBRAIC FAMILIES

Constant

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A rule that returns the same value no matter what the input is.
Memory jogger: Same output every time.

FORMULA AND BEHAVIOR

```
f(x) = c
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Step, indicator, baseline, intercept

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Constant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Identity

DIFFICULTY: INTERMEDIATE

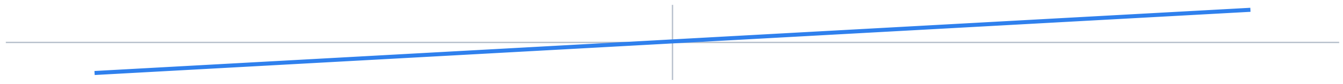
MEANING AND MEMORY

Meaning: A rule that sends every input back to itself unchanged.
Memory jogger: Input passes through untouched.

FORMULA AND BEHAVIOR

```
f(x) = x
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

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Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Linear, inverse, residual, pass-through transform

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Identity without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Quadratic

DIFFICULTY: INTERMEDIATE

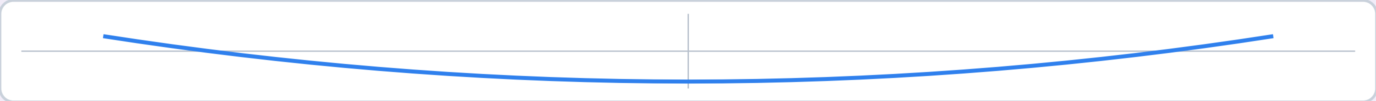
MEANING AND MEMORY

Meaning: A second-degree polynomial whose graph bends into a parabola.
Memory jogger: U-shape or inverted U-shape.

FORMULA AND BEHAVIOR

```
f(x) = a*x^2 + b*x + c
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: market impact can be modeled as cost = a*size + b*size^2 when large trades become increasingly expensive.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, parabola, convexity, transaction cost

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Quadratic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cubic

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A third-degree polynomial that can express asymmetric curvature and inflection.
Memory jogger: S-curve with polynomial tails.

FORMULA AND BEHAVIOR

```
f(x) = a*x^3 + b*x^2 + c*x + d
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, inflection, skew, nonlinear response

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cubic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Quartic

DIFFICULTY: INTERMEDIATE

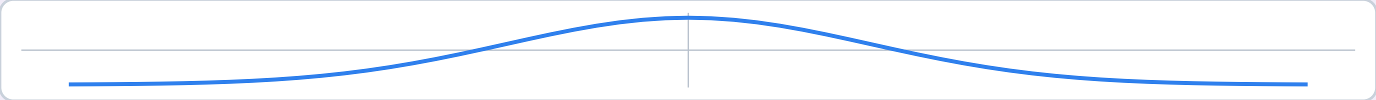
MEANING AND MEMORY

Meaning: A fourth-degree polynomial with enough curvature to form multiple turning points.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = a*x^4 + b*x^3 + c*x^2 + d*x + e
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, curvature, tails, smoothing

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Quartic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Quintic

DIFFICULTY: INTERMEDIATE

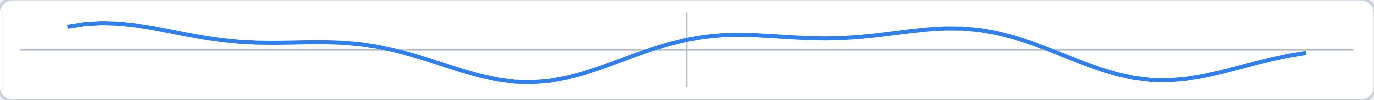
MEANING AND MEMORY

Meaning: A fifth-degree polynomial that can produce complex curvature but usually lacks a general radical solution.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = a*x^5 + ... + k
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, spline, interpolation, numerical roots

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Quintic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Polynomial

DIFFICULTY: INTERMEDIATE

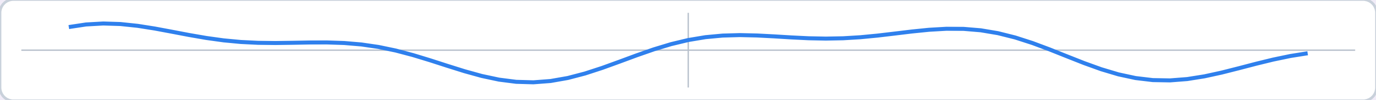
MEANING AND MEMORY

Meaning: A finite sum of powers of x with coefficients.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = sum_{k=0}^n a_k*x^k
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

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Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: High-degree polynomials can fit noise extremely well; smooth curves are not automatically robust signals.
Related terms: Monomial, power, regression basis, Taylor approximation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Polynomial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Monomial

DIFFICULTY: INTERMEDIATE

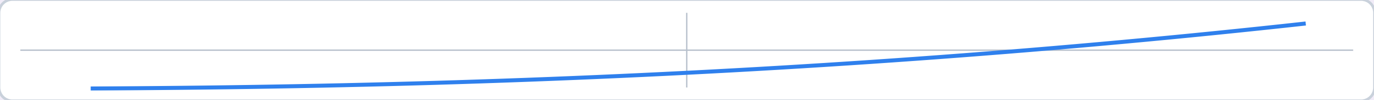
MEANING AND MEMORY

Meaning: A single polynomial term.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = c*x^n
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, power, basis expansion

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Monomial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Binomial

DIFFICULTY: INTERMEDIATE

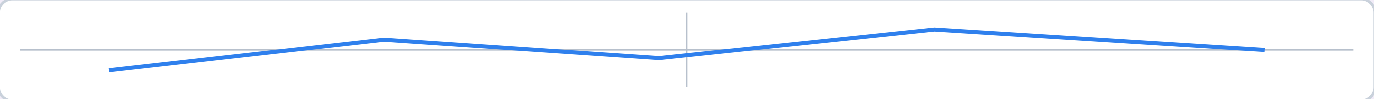
MEANING AND MEMORY

Meaning: A two-term algebraic expression, often used as a building block for expansions.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = term_1 + term_2
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

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Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, expansion, binomial distribution

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Binomial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Trinomial

DIFFICULTY: INTERMEDIATE

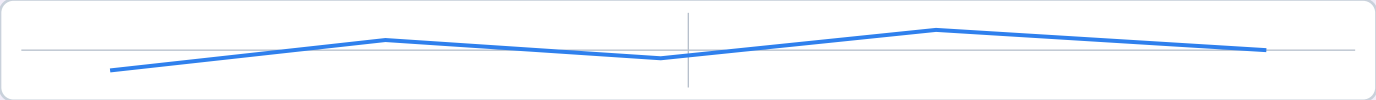
MEANING AND MEMORY

Meaning: A three-term algebraic expression, commonly seen in quadratics.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = term_1 + term_2 + term_3
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Quadratic, polynomial, factoring

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Trinomial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

CORE ALGEBRAIC FAMILIES

Power

DIFFICULTY: INTERMEDIATE

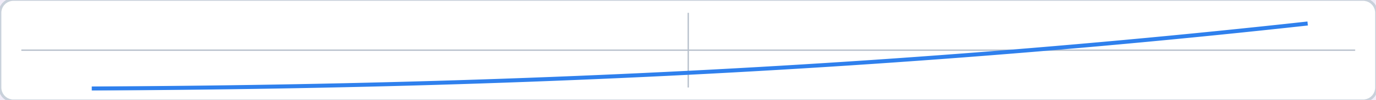
MEANING AND MEMORY

Meaning: A function where the input is raised to a fixed exponent.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = a*x^p
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
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SYSTEM FIT AND IMPLEMENTATION

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TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Root, monomial, scaling law, elasticity

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Power without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Root

DIFFICULTY: INTERMEDIATE

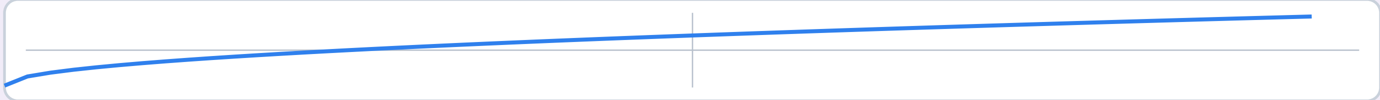
MEANING AND MEMORY

Meaning: A function that reverses a power operation.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = x^(1/n)
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

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Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Power, radical, variance scaling

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Root without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Square root

DIFFICULTY: INTERMEDIATE

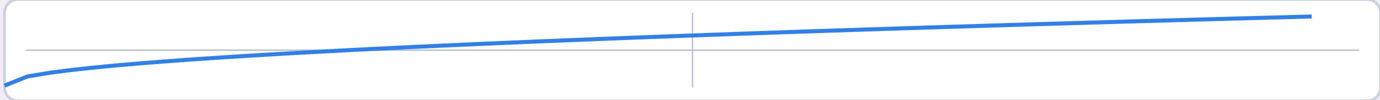
MEANING AND MEMORY

Meaning: The nonnegative value whose square equals the input.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = sqrt(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: annualized_vol = daily_vol*sqrt(252), using square-root time scaling as an approximation.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Root, volatility, standard deviation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Square root without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cube root

DIFFICULTY: INTERMEDIATE

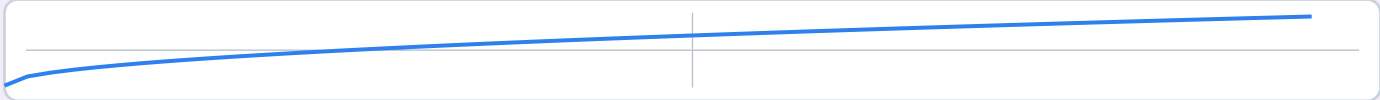
MEANING AND MEMORY

Meaning: The value whose cube equals the input, defined for positive and negative real inputs.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = cbrt(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Root, signed scale transform

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cube root without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Reciprocal

DIFFICULTY: INTERMEDIATE

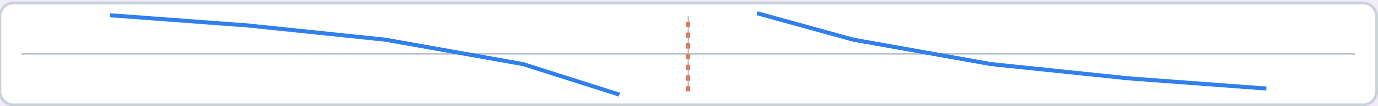
MEANING AND MEMORY

Meaning: A function that maps a value to its multiplicative inverse.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = 1/x
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Rational, asymptote, inverse proportion

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Reciprocal without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Rational

DIFFICULTY: INTERMEDIATE

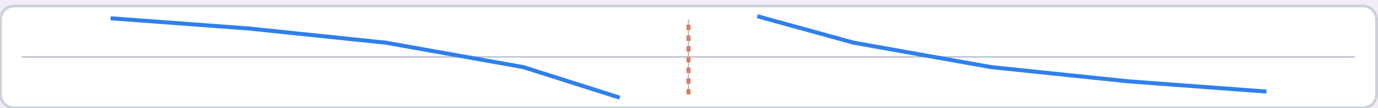
MEANING AND MEMORY

Meaning: A ratio of two polynomials.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = P(x)/Q(x), Q(x) != 0
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Reciprocal, asymptote, transfer function

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Rational without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Radical

DIFFICULTY: INTERMEDIATE

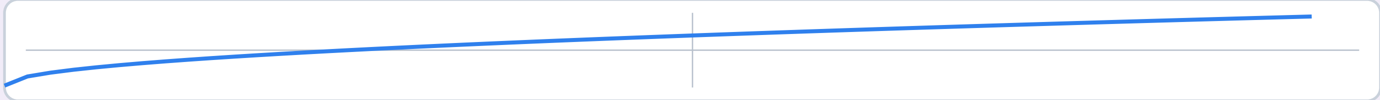
MEANING AND MEMORY

Meaning: An expression or function containing roots.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
f(x) = expression containing roots
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Root, algebraic, domain restriction

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Radical without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Algebraic

DIFFICULTY: INTERMEDIATE

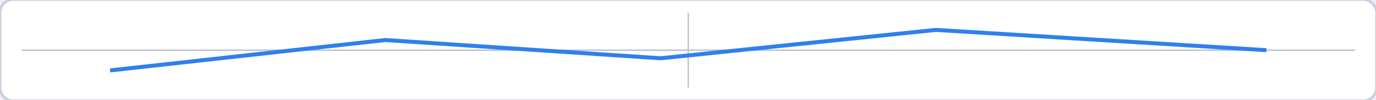
MEANING AND MEMORY

Meaning: A function defined by polynomial equations and finite algebraic operations.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
F(x, f(x)) = 0 for some polynomial F
```

Formula intuition: Behavior is controlled by degree, exponent, coefficients, and domain restrictions. Small coefficient changes can alter curvature and roots.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Polynomial, radical, rational, implicit

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Algebraic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Transcendental

DIFFICULTY: INTERMEDIATE

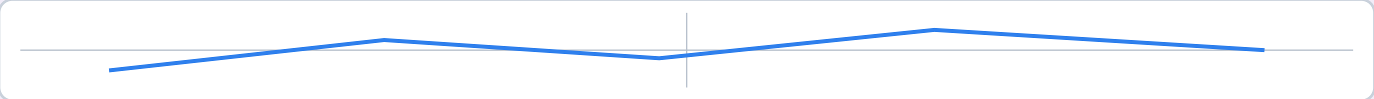
MEANING AND MEMORY

Meaning: A function that cannot be captured by finite polynomial equations alone.
Memory jogger: Think: curve vocabulary for simple model shape.

FORMULA AND BEHAVIOR

```
not the root of a polynomial equation in x and f(x)
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to design feature transforms, regression curves, cost models, response surfaces, and simple approximations before moving to heavier models.
Simple market example: Example: transform volatility, volume, or spread into a smoother explanatory curve before testing predictive value.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Exponential, logarithmic, trigonometric, special functions

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Transcendental without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 02

Piecewise, Discrete, and Threshold Functions

Piecewise and threshold functions turn continuous values into regimes, buckets, trades, gates, and execution logic.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Piecewise

DIFFICULTY: INTERMEDIATE

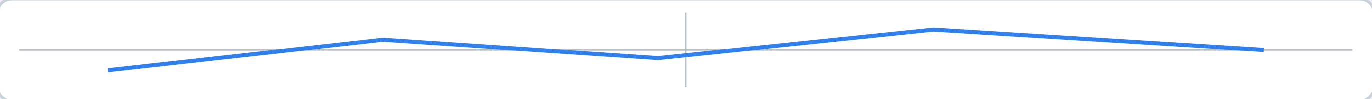
MEANING AND MEMORY

Meaning: A function assembled from different rules on different regions of the input space.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
f(x) = rule_A if condition_A, rule_B otherwise
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Step, threshold, ReLU, spline

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Piecewise without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Step

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function that stays flat, then jumps to a new level at breakpoints.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
f(x) changes value at breakpoints
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Heaviside, indicator, threshold, floor

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Step without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Floor

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A rounding function that always moves down to the nearest integer.
Memory jogger: Round down.

FORMULA AND BEHAVIOR

```
floor(x) = largest integer <= x
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: floor(price/tick_size)*tick_size maps a theoretical price onto an executable tick grid.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Floor is not truncation for negative numbers; floor(-1.2) is -2.
Related terms: Greatest integer, fractional part, bucketization

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Floor without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Ceiling

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A rounding function that always moves up to the nearest integer.
Memory jogger: Round up.

FORMULA AND BEHAVIOR

```
ceil(x) = smallest integer >= x
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Ceiling moves upward, which can increase risk if used to round position size.
Related terms: Floor, rounding, tick size

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Ceiling without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Greatest integer

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: Another name for the floor function.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
[x] = floor(x)
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: floor(price/tick_size)*tick_size maps a theoretical price onto an executable tick grid.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Floor, step, bucket

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Greatest integer without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Nearest integer

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A rounding function that maps to the closest integer.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
round(x) = integer closest to x
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Floor, ceiling, quantization

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Nearest integer without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Absolute value

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A distance-from-zero function that removes sign.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
abs(x) = x if x >= 0, -x if x < 0
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Norm, distance, L1 loss, kink

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Absolute value without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sign

DIFFICULTY: INTERMEDIATE

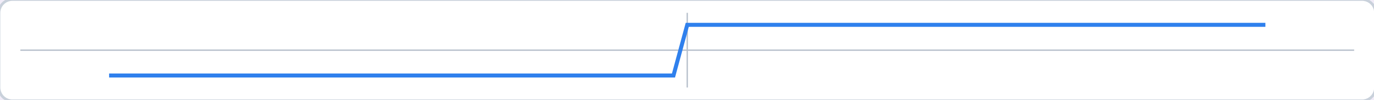
MEANING AND MEMORY

Meaning: A function that keeps only direction: negative, zero, or positive.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
sign(x) = -1, 0, or 1
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement as a vectorized boolean mask or clipped array. Log the threshold, data timestamp, and whether the rule was fit in-sample.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Indicator, threshold, direction signal

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sign without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Indicator

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A binary membership test written as a function.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
1_A(x) = 1 if x in A, else 0
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: 1_{spread < max_spread} gates a strategy so it only trades when liquidity is acceptable.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement as a vectorized boolean mask or clipped array. Log the threshold, data timestamp, and whether the rule was fit in-sample.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A binary mask can hide how close observations were to the decision boundary.
Related terms: Step, boolean mask, event condition

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Indicator without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Heaviside

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A canonical step function for switching a process on after a boundary.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
H(x) = 0 for x < 0, 1 for x >= 0
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement as a vectorized boolean mask or clipped array. Log the threshold, data timestamp, and whether the rule was fit in-sample.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: The value at zero is convention-dependent; define it explicitly.
Related terms: Step, impulse response, threshold

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Heaviside without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Ramp

DIFFICULTY: INTERMEDIATE

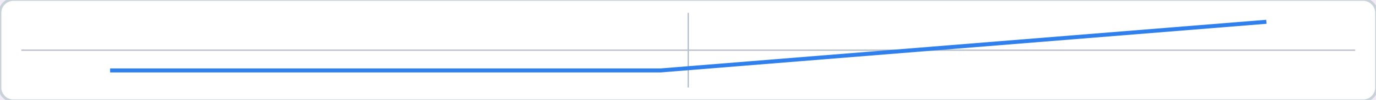
MEANING AND MEMORY

Meaning: A zero-below-threshold and linear-above-threshold function.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
R(x) = max(0, x)
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: ReLU, hinge loss, piecewise linear

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Ramp without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Clamp

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function that caps values inside a lower and upper bound.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
clamp(x,a,b)=min(max(x,a),b)
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: clamp(position, -max_risk, max_risk) prevents a sizing rule from exceeding risk limits.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement as a vectorized boolean mask or clipped array. Log the threshold, data timestamp, and whether the rule was fit in-sample.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Winsorization, bounds, saturation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Clamp without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Threshold

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A cutoff rule that changes behavior once a score crosses a level.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
signal = 1 if score > tau, else 0
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement as a vectorized boolean mask or clipped array. Log the threshold, data timestamp, and whether the rule was fit in-sample.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Indicator, sign, decision rule

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Threshold without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Modulo

DIFFICULTY: INTERMEDIATE

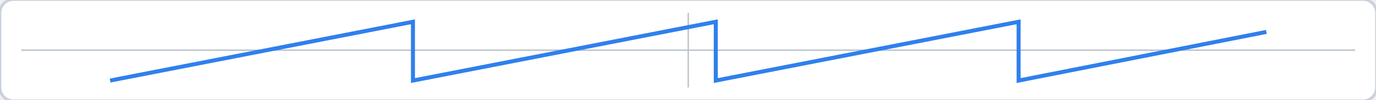
MEANING AND MEMORY

Meaning: A remainder function that wraps values around a fixed period.
Memory jogger: Wrap around after a fixed length.

FORMULA AND BEHAVIOR

```
x mod m = remainder after division by m
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: minute_of_day mod 390 wraps intraday observations into a repeatable trading-session cycle.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Different languages handle negative modulo differently; test before using cyclic features.
Related terms: Periodic, fractional part, cyclic feature

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Modulo without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Fractional part

DIFFICULTY: INTERMEDIATE

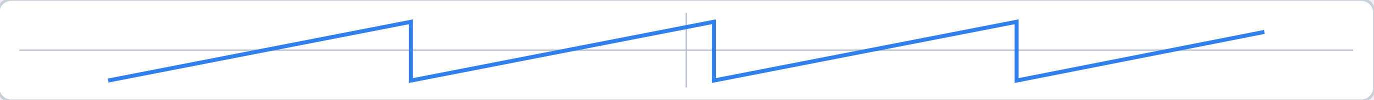
MEANING AND MEMORY

Meaning: The portion of x left after subtracting its integer floor.
Memory jogger: Keep only what is after the decimal step.

FORMULA AND BEHAVIOR

```
{x} = x - floor(x)
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Implement with vectorized numpy operations. Unit-test negative values, exact integers, NaN handling, and tick-size rounding edge cases.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Modulo, floor, sawtooth wave

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Fractional part without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Dirichlet

DIFFICULTY: INTERMEDIATE

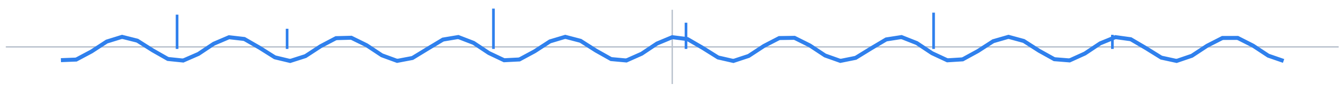
MEANING AND MEMORY

Meaning: A pathological function that separates rational and irrational inputs everywhere.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
D(x) = 1 if x is rational, 0 if irrational
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Discontinuous, rational/irrational, pathology

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Dirichlet without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Thomae's

DIFFICULTY: INTERMEDIATE

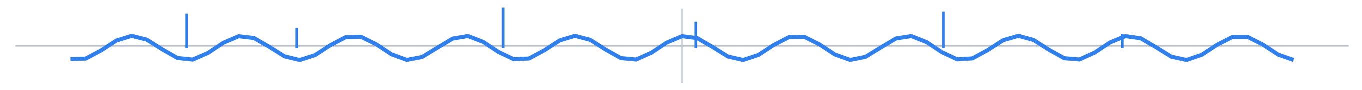
MEANING AND MEMORY

Meaning: A function that is spiky at rationals and zero at irrationals, showing subtle continuity behavior.
Memory jogger: Think: rule switches, buckets, and hard boundaries.

FORMULA AND BEHAVIOR

```
T(p/q)=1/q for rational p/q in lowest terms; T(x)=0 if irrational
```

Formula intuition: The boundary matters. Most errors happen at exact cutoffs, negative values, missing data, or values just barely across the threshold.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to express rules: entries, exits, filters, market regimes, tick-size buckets, hard caps, and boolean masks.
Simple market example: Example: score > threshold becomes a trade signal; score <= threshold becomes flat or no-trade.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Continuity, rationals, removable-like spikes

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Thomae's without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 03

Exponential, Logarithmic, and Saturating Functions

Exponential and logarithmic functions explain compounding, decay, scaling, half-life, and probability squashing.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Exponential

DIFFICULTY: INTERMEDIATE

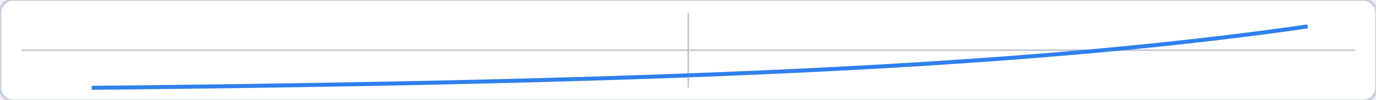
MEANING AND MEMORY

Meaning: A function where equal input changes multiply the output by a constant factor.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(x)=a*b^x
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: use a decayed average so the model adapts faster to recent market behavior without forgetting history instantly.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Natural exponential, growth, decay, compounding

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Exponential without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Natural exponential

DIFFICULTY: INTERMEDIATE

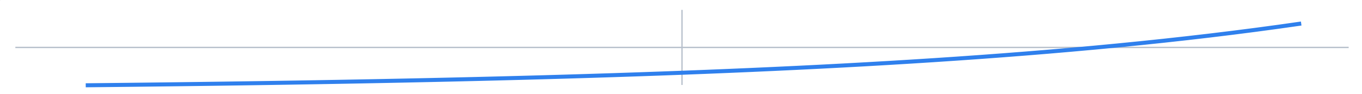
MEANING AND MEMORY

Meaning: The base-e exponential, central to continuous compounding and differential equations.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(x)=e^x
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: use a decayed average so the model adapts faster to recent market behavior without forgetting history instantly.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Logarithm, continuous compounding, differential equations

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Natural exponential without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Exponential growth

DIFFICULTY: INTERMEDIATE

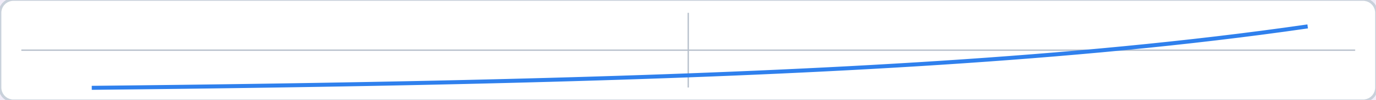
MEANING AND MEMORY

Meaning: A process that grows in proportion to its current size.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(t)=A*e^(k*t), k>0
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: use a decayed average so the model adapts faster to recent market behavior without forgetting history instantly.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Compounding, trend, instability

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Exponential growth without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Exponential decay

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A process that shrinks in proportion to its current size.
Memory jogger: Old information fades by a fixed percentage.

FORMULA AND BEHAVIOR

```
f(t)=A*e^(-lambda*t), lambda>0
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: an EWMA volatility estimator gives yesterday more weight than last month using decay by half-life.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: EWMA, half-life, forgetting factor

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Exponential decay without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Logarithmic

DIFFICULTY: INTERMEDIATE

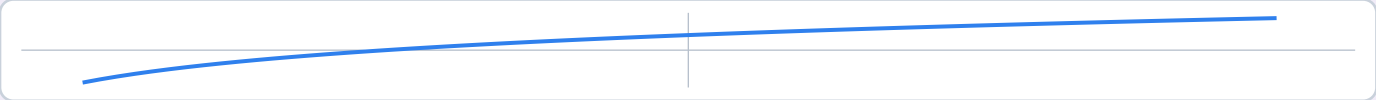
MEANING AND MEMORY

Meaning: The inverse of an exponential function.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(x)=log_b(x)
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: log_return = ln(P_t/P_{t-1}), which adds cleanly across time windows.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A logarithm requires positive input unless you deliberately use a signed or shifted transform.
Related terms: Exponential, log return, scale compression

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Logarithmic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Natural logarithm

DIFFICULTY: INTERMEDIATE

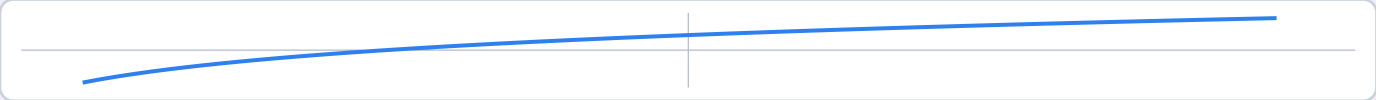
MEANING AND MEMORY

Meaning: The base-e logarithm, central to log returns and continuous growth.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(x)=ln(x)
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: log_return = ln(P_t/P_{t-1}), which adds cleanly across time windows.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not mix arithmetic returns and log returns without knowing which aggregation rule you need.
Related terms: Log return, likelihood, entropy

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Natural logarithm without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Common logarithm

DIFFICULTY: INTERMEDIATE

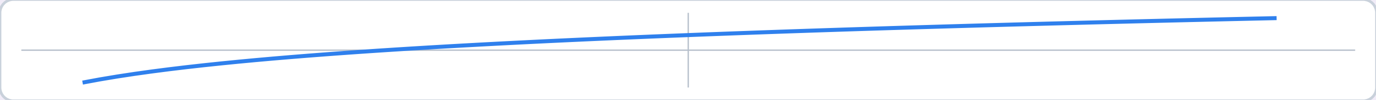
MEANING AND MEMORY

Meaning: The base-10 logarithm, useful for orders of magnitude.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
f(x)=log_10(x)
```

Formula intuition: These functions usually change scale nonlinearly; they are excellent for compression, growth, decay, and inverse-compounding logic.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: use a decayed average so the model adapts faster to recent market behavior without forgetting history instantly.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Orders of magnitude, log-scale plots

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Common logarithm without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Logistic

DIFFICULTY: INTERMEDIATE

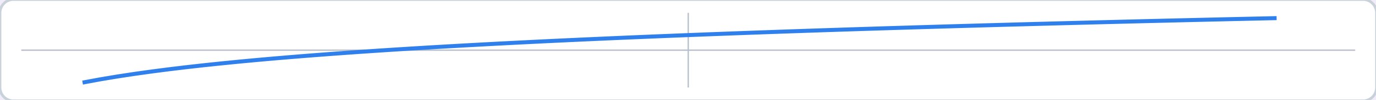
MEANING AND MEMORY

Meaning: An S-shaped curve that grows quickly then saturates at an upper limit.
Memory jogger: Fast middle, flat ends.

FORMULA AND BEHAVIOR

```
f(x)=L/(1+e^(-k*(x-x0)))
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: map a model score of -infinity to +infinity into a probability-like value between 0 and 1.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use numerically stable forms: subtract max before softmax and avoid overflow in exp for large positive or negative scores.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sigmoid, probability, saturation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Logistic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sigmoid

DIFFICULTY: INTERMEDIATE

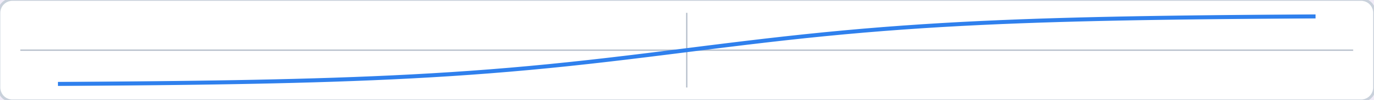
MEANING AND MEMORY

Meaning: A standard S-curve mapping real numbers into the interval (0,1).
Memory jogger: Squash any score into 0 to 1.

FORMULA AND BEHAVIOR

```
sigma(x)=1/(1+e^(-x))
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: map a model score of -infinity to +infinity into a probability-like value between 0 and 1.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use numerically stable forms: subtract max before softmax and avoid overflow in exp for large positive or negative scores.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A sigmoid output is probability-like only if the model is calibrated.
Related terms: Logistic, activation, classifier probability

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sigmoid without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Softplus

DIFFICULTY: INTERMEDIATE

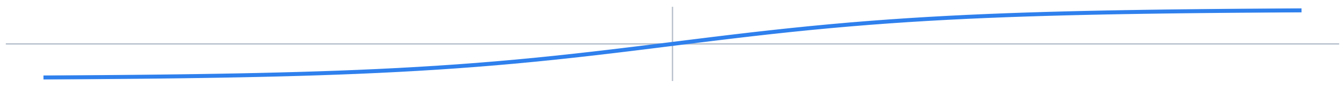
MEANING AND MEMORY

Meaning: A smooth approximation to ReLU that never has a hard kink.
Memory jogger: Think: compounding, compression, or saturation.

FORMULA AND BEHAVIOR

```
softplus(x)=ln(1+e^x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for compounding, log returns, EWMA decay, probability maps, bounded scores, and nonlinear scaling of noisy inputs.
Simple market example: Example: use a decayed average so the model adapts faster to recent market behavior without forgetting history instantly.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use numerically stable forms: subtract max before softmax and avoid overflow in exp for large positive or negative scores.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: ReLU, smooth approximation, convex transform

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Softplus without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 04

Trigonometric and Inverse Trigonometric Functions

Trigonometric functions are the natural language of cycles, phase, seasonality, Fourier features, and oscillation.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Sine

DIFFICULTY: INTERMEDIATE

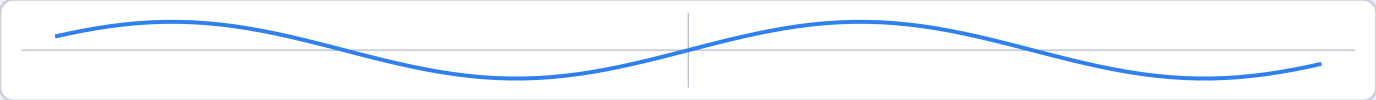
MEANING AND MEMORY

Meaning: A smooth periodic wave built from circular motion.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
sin(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: encode time-of-day as sin(2*pi*t/390) and cos(2*pi*t/390) so open and close cycles are smooth.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Cosine, periodic, Fourier feature

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cosine

DIFFICULTY: INTERMEDIATE

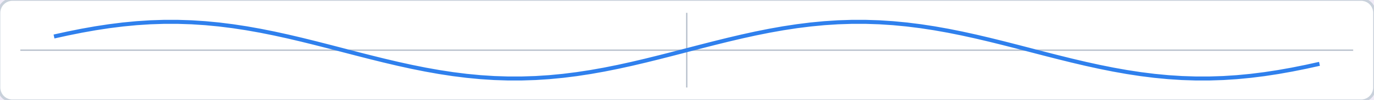
MEANING AND MEMORY

Meaning: A sine wave shifted by a quarter cycle, also built from circular motion.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
cos(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: encode time-of-day as sin(2*pi*t/390) and cos(2*pi*t/390) so open and close cycles are smooth.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sine, phase, Fourier feature

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cosine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Tangent

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A periodic ratio of sine to cosine with vertical asymptotes.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
tan(x)=sin(x)/cos(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Asymptotes create huge values near undefined points; clip or encode safely.
Related terms: Sine, cosine, asymptote

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Tangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cotangent

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: The reciprocal of tangent.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
cot(x)=cos(x)/sin(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Tangent, reciprocal trig

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cotangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Secant

DIFFICULTY: INTERMEDIATE

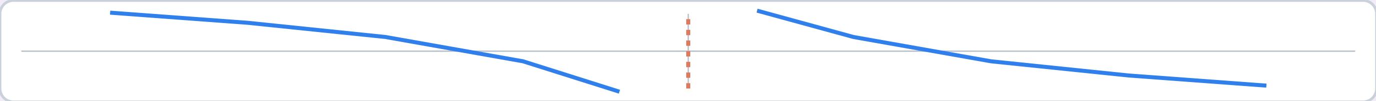
MEANING AND MEMORY

Meaning: The reciprocal of cosine.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
sec(x)=1/cos(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Cosine, reciprocal trig

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Secant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cosecant

DIFFICULTY: INTERMEDIATE

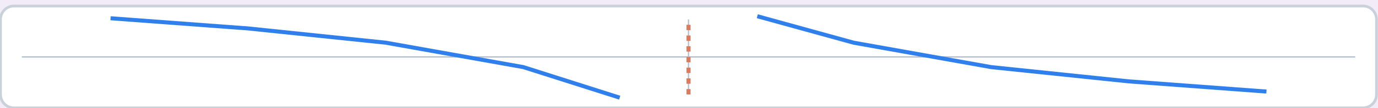
MEANING AND MEMORY

Meaning: The reciprocal of sine.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
csc(x)=1/sin(x)
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sine, reciprocal trig

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cosecant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arcsine

DIFFICULTY: INTERMEDIATE

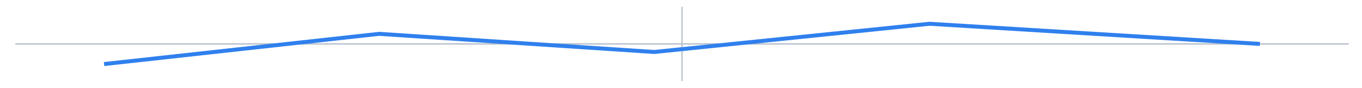
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a sine value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arcsin(x) returns angle with sine x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sine, inverse, restricted domain

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arcsine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arccosine

DIFFICULTY: INTERMEDIATE

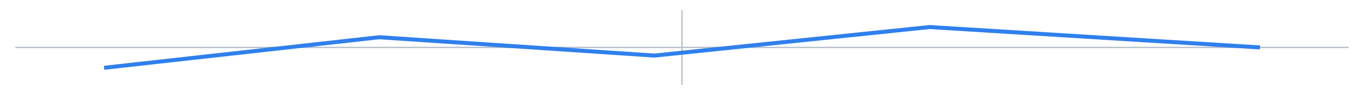
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a cosine value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arccos(x) returns angle with cosine x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Cosine, inverse, restricted domain

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arccosine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arctangent

DIFFICULTY: INTERMEDIATE

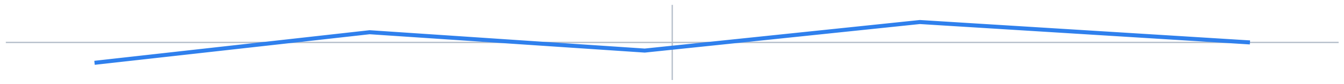
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a tangent value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arctan(x) returns angle with tangent x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Tangent, inverse, bounded angle

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arctangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arccotangent

DIFFICULTY: INTERMEDIATE

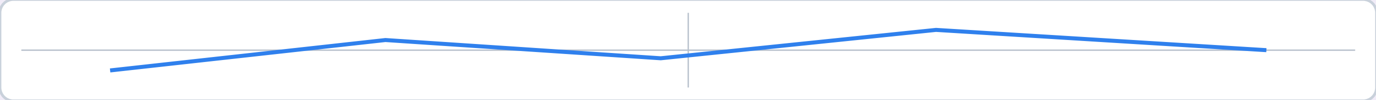
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a cotangent value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arccot(x) returns angle with cotangent x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Conventions differ for output range; document the branch.
Related terms: Cotangent, inverse, convention

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arccotangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arcsecant

DIFFICULTY: INTERMEDIATE

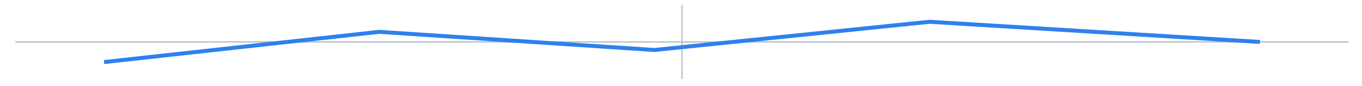
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a secant value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arcsec(x) returns angle with secant x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Domain excludes (-1,1); validate inputs.
Related terms: Secant, inverse, domain gap

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arcsecant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arccosecant

DIFFICULTY: INTERMEDIATE

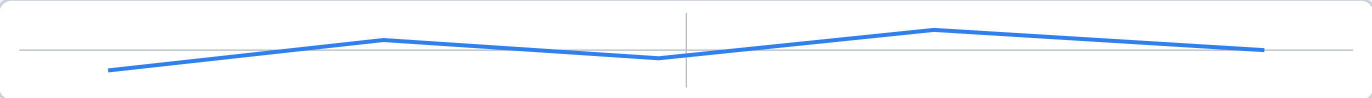
MEANING AND MEMORY

Meaning: The inverse relation that recovers an angle from a cosecant value.
Memory jogger: Think: angle, cycle, and phase.

FORMULA AND BEHAVIOR

```
arccsc(x) returns angle with cosecant x
```

Formula intuition: Angles can be in radians or degrees; numerical libraries usually expect radians. Phase shifts move the cycle without changing its shape.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to encode cycles, phase, seasonality, Fourier-style features, and repeated calendar or intraday structure.
Simple market example: Example: model a repeating intraday liquidity cycle without creating an artificial discontinuity at the session boundary.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Domain excludes (-1,1); validate inputs.
Related terms: Cosecant, inverse, domain gap

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arccosecant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 05

Hyperbolic Functions

Hyperbolic functions are exponential-derived curves that appear in smooth saturation, transforms, and differential-equation systems.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Hyperbolic sine

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: An exponential-based analog of sine.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
sinh(x)=(e^x-e^-x)/2
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Exponential, asinh, signed growth

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic sine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Hyperbolic cosine

DIFFICULTY: INTERMEDIATE

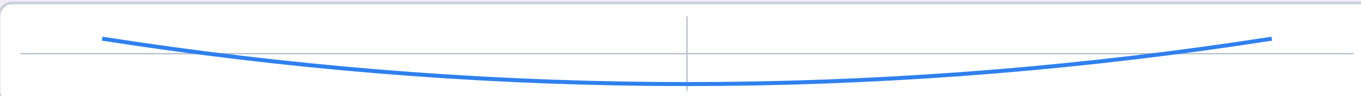
MEANING AND MEMORY

Meaning: An exponential-based analog of cosine with a U-shaped profile.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
cosh(x)=(e^x+e^-x)/2
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Exponential, sech, catenary shape

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic cosine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Hyperbolic tangent

DIFFICULTY: INTERMEDIATE

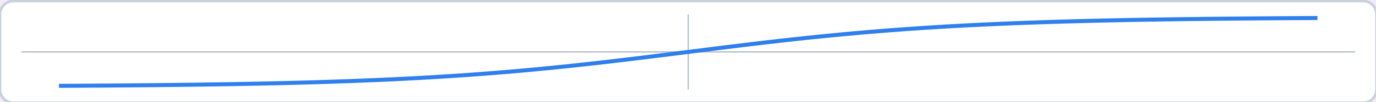
MEANING AND MEMORY

Meaning: A smooth bounded S-curve mapping real inputs into (-1,1).
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
tanh(x)=sinh(x)/cosh(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sigmoid, tanh activation, bounded signal

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic tangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Hyperbolic cotangent

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: The reciprocal of hyperbolic tangent.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
coth(x)=cosh(x)/sinh(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Tanh, reciprocal, asymptote

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic cotangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Hyperbolic secant

DIFFICULTY: INTERMEDIATE

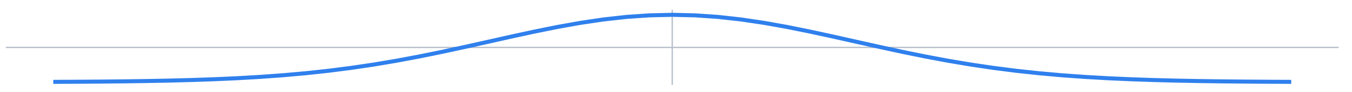
MEANING AND MEMORY

Meaning: The reciprocal of hyperbolic cosine, often forming a bell-shaped curve.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
sech(x)=1/cosh(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Cosh, bell shape, density family

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic secant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Hyperbolic cosecant

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: The reciprocal of hyperbolic sine.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
csch(x)=1/sinh(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sinh, reciprocal, singularity

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Hyperbolic cosecant without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Inverse hyperbolic sine

DIFFICULTY: INTERMEDIATE

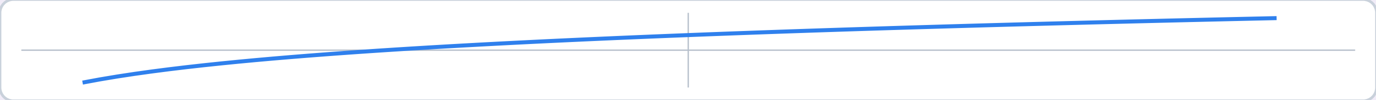
MEANING AND MEMORY

Meaning: The inverse of sinh, useful because it acts log-like for large values while handling zero and negatives.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
asinh(x)=ln(x+sqrt(x^2+1))
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Asinh transform, log-like scaling

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Inverse hyperbolic sine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Inverse hyperbolic cosine

DIFFICULTY: INTERMEDIATE

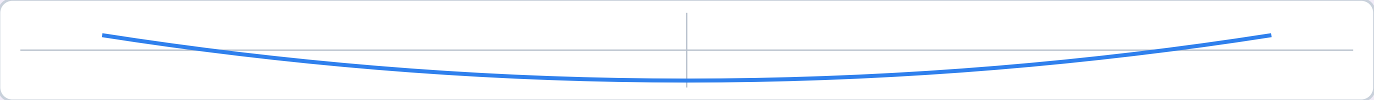
MEANING AND MEMORY

Meaning: The inverse of cosh, defined for x >= 1.
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
acosh(x)=ln(x+sqrt(x^2-1)), x>=1
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: acosh is defined for x >= 1 in real arithmetic.
Related terms: Cosh inverse, domain restriction

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Inverse hyperbolic cosine without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Inverse hyperbolic tangent

DIFFICULTY: INTERMEDIATE

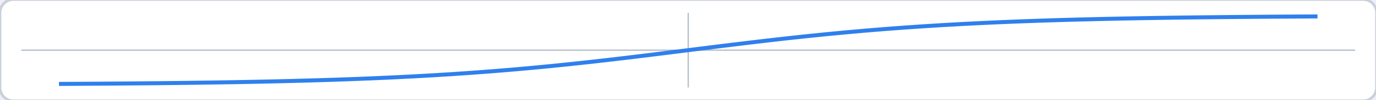
MEANING AND MEMORY

Meaning: The inverse of tanh, defined on (-1,1).
Memory jogger: Think: exponential geometry and smooth saturation.

FORMULA AND BEHAVIOR

```
atanh(x)=0.5*ln((1+x)/(1-x)), |x|<1
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it for smooth signed transforms, bounded signals, log-like scaling, and nonlinear activation behavior.
Simple market example: Example: squash an unbounded alpha score into a controlled range before converting it into position size.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: atanh only accepts |x| < 1 for real output.
Related terms: Tanh inverse, Fisher transform

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Inverse hyperbolic tangent without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 06

Periodic, Oscillatory, and Wave Functions

Wave functions model repeating structure, cyclic time features, intraday rhythms, and frequency-domain decompositions.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Periodic

DIFFICULTY: INTERMEDIATE

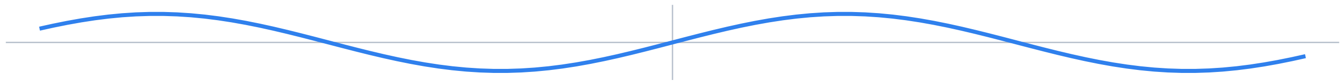
MEANING AND MEMORY

Meaning: A function that repeats after a fixed interval.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
f(x+T)=f(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: encode time-of-day as sin(2*pi*t/390) and cos(2*pi*t/390) so open and close cycles are smooth.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sine, modulo, seasonality

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Periodic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Oscillatory

DIFFICULTY: INTERMEDIATE

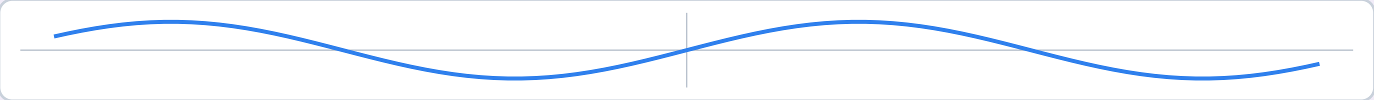
MEANING AND MEMORY

Meaning: A function that repeatedly moves around a reference level.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
values repeatedly move around a reference level
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: generate synthetic cyclical returns to test whether a pipeline can detect seasonality without leaking future data.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Periodic, wave, mean reversion

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Oscillatory without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sinusoidal

DIFFICULTY: INTERMEDIATE

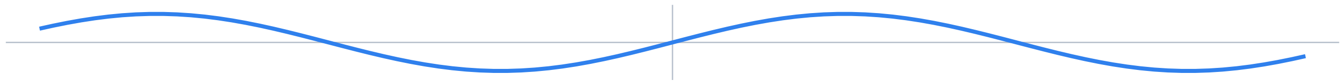
MEANING AND MEMORY

Meaning: A sine or cosine shaped periodic function.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
f(t)=A*sin(omega*t+phi)+c
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: encode time-of-day as sin(2*pi*t/390) and cos(2*pi*t/390) so open and close cycles are smooth.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sine, cosine, Fourier

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sinusoidal without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Wave

DIFFICULTY: INTERMEDIATE

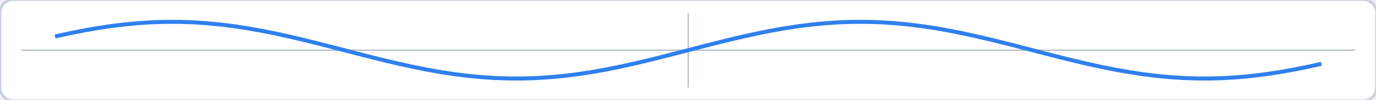
MEANING AND MEMORY

Meaning: A repeated pattern described by amplitude, frequency, phase, and shape.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
amplitude, frequency, phase, and shape define repeating motion
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: encode time-of-day as sin(2*pi*t/390) and cos(2*pi*t/390) so open and close cycles are smooth.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sinusoidal, square wave, frequency

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Wave without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Square wave

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A periodic function that jumps between two levels.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
alternates between high and low levels
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: generate synthetic cyclical returns to test whether a pipeline can detect seasonality without leaking future data.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Step, threshold, Fourier series

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Square wave without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Triangle wave

DIFFICULTY: INTERMEDIATE

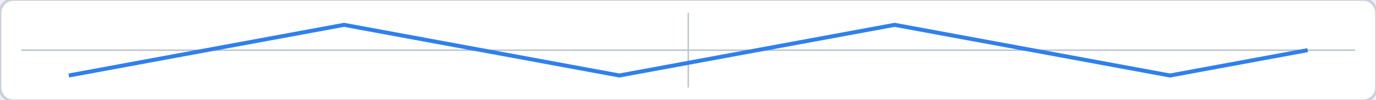
MEANING AND MEMORY

Meaning: A periodic wave with straight-line rises and falls.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
piecewise linear rise and fall
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: generate synthetic cyclical returns to test whether a pipeline can detect seasonality without leaking future data.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Piecewise linear, absolute value, periodic

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Triangle wave without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sawtooth wave

DIFFICULTY: INTERMEDIATE

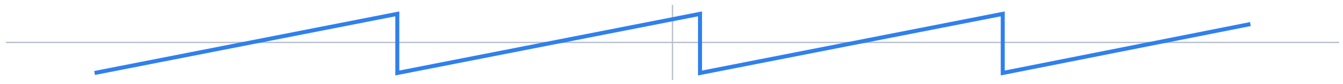
MEANING AND MEMORY

Meaning: A periodic ramp that resets sharply.
Memory jogger: Think: repeated motion through time.

FORMULA AND BEHAVIOR

```
linear ramp followed by reset
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model repeated market rhythms, seasonality, cyclical demand, volatility pulses, and synthetic test signals.
Simple market example: Example: generate synthetic cyclical returns to test whether a pipeline can detect seasonality without leaking future data.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Modulo, fractional part, periodic reset

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sawtooth wave without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 07

Function Properties and Mapping Behavior

Function properties tell you whether a transform preserves ordering, loses information, can be inverted, or covers a target range.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Even

DIFFICULTY: INTERMEDIATE

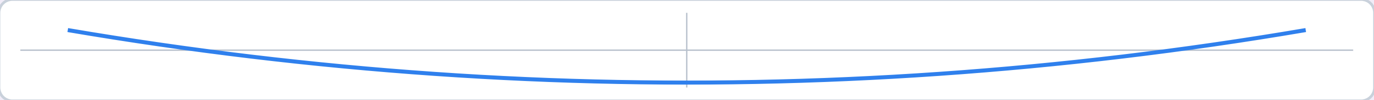
MEANING AND MEMORY

Meaning: A function symmetric around the vertical axis.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
f(-x)=f(x)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Symmetry, cosine, Gaussian

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Even without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Odd

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function symmetric through the origin.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
f(-x)=-f(x)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Symmetry, sine, tanh

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Odd without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Increasing

DIFFICULTY: INTERMEDIATE

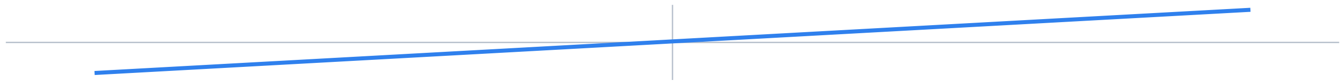
MEANING AND MEMORY

Meaning: A function whose output never falls as input rises.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
x1 < x2 implies f(x1) <= f(x2)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Monotonic, ranking, invertibility

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Increasing without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

FUNCTION PROPERTIES AND MAPPING BEHAVIOR

Decreasing

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function whose output never rises as input rises.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
x1 < x2 implies f(x1) >= f(x2)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Monotonic, inverse relationship

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Decreasing without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Monotonic

DIFFICULTY: INTERMEDIATE

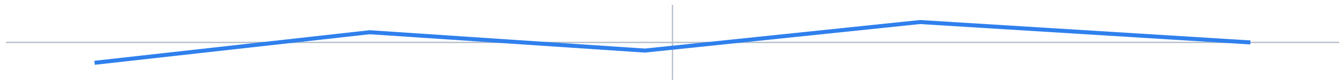
MEANING AND MEMORY

Meaning: A function that moves in only one direction, allowing flat regions.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
function is entirely nondecreasing or nonincreasing
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Increasing, decreasing, order preservation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Monotonic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Strictly increasing

DIFFICULTY: INTERMEDIATE

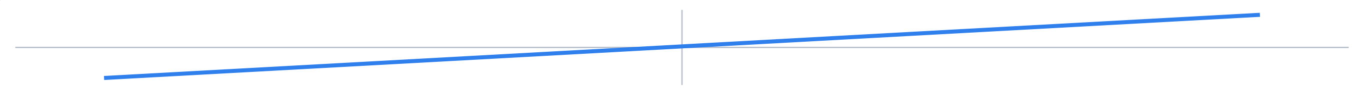
MEANING AND MEMORY

Meaning: A function whose output always rises when input rises.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
x1 < x2 implies f(x1) < f(x2)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Injective, invertible, ranking

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Strictly increasing without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

FUNCTION PROPERTIES AND MAPPING BEHAVIOR

Strictly decreasing

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function whose output always falls when input rises.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
x1 < x2 implies f(x1) > f(x2)
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Injective, invertible, ranking

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Strictly decreasing without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

One-to-one

DIFFICULTY: INTERMEDIATE

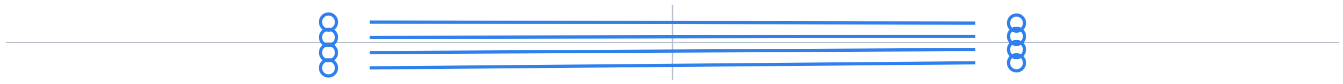
MEANING AND MEMORY

Meaning: A mapping where each output comes from at most one input.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
different inputs produce different outputs
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Injective, invertible, inverse

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of One-to-one without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Many-to-one

DIFFICULTY: INTERMEDIATE

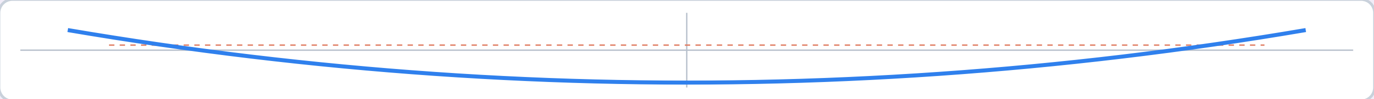
MEANING AND MEMORY

Meaning: A mapping where multiple inputs can collapse to the same output.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
multiple inputs can share the same output
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Lossy transform, aggregation, squaring

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Many-to-one without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Onto

DIFFICULTY: INTERMEDIATE

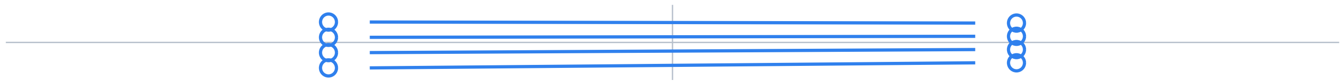
MEANING AND MEMORY

Meaning: A mapping that covers every value in the target set.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
every target output is reached
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Surjective, codomain, coverage

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Onto without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Bijective

DIFFICULTY: INTERMEDIATE

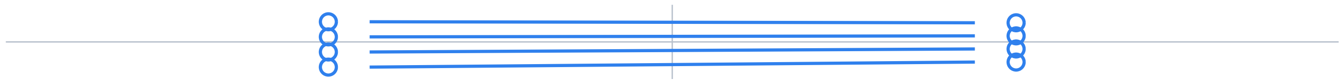
MEANING AND MEMORY

Meaning: A mapping that is both one-to-one and onto.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
injective and surjective
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Injective, surjective, inverse

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Bijective without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Injective

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A formal name for one-to-one.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
one-to-one
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: One-to-one, inverse, uniqueness

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Injective without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Surjective

DIFFICULTY: INTERMEDIATE

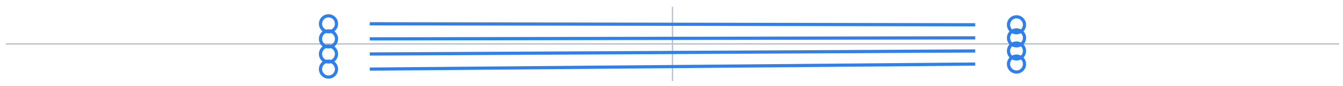
MEANING AND MEMORY

Meaning: A formal name for onto.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
onto
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Onto, codomain, coverage

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Surjective without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Invertible

DIFFICULTY: INTERMEDIATE

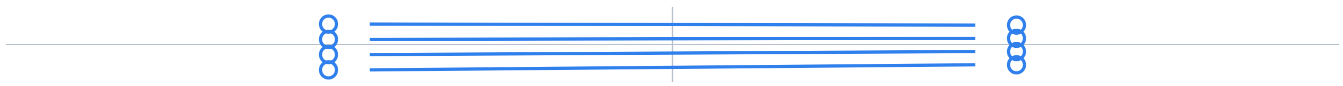
MEANING AND MEMORY

Meaning: A function whose input can be recovered from its output.
Memory jogger: Think: what the mapping preserves or destroys.

FORMULA AND BEHAVIOR

```
there exists g such that g(f(x))=x
```

Formula intuition: This is a property of a function plus its stated domain and target set, not just the formula printed on a page.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it as a diagnostic for whether a transform preserves ranking, destroys information, supports inversion, or changes the target range.
Simple market example: Example: check whether a volatility transform preserves order; if not, rankings may change and backtest interpretation changes.

SYSTEM FIT AND IMPLEMENTATION

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Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Always specify domain, codomain, and range. Many mapping properties change when the domain is restricted.
Related terms: Inverse, bijective, domain restriction

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Invertible without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 08

Continuity, Discontinuity, and Smoothness

Continuity and smoothness determine whether a model is optimizer-friendly, simulation-stable, and interpretable around boundaries.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Continuous

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function with no holes, jumps, or breaks at the point under discussion.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
lim_{x->a} f(x) = f(a)
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Limit, smoothness, differentiability

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Continuous without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Discontinuous

DIFFICULTY: INTERMEDIATE

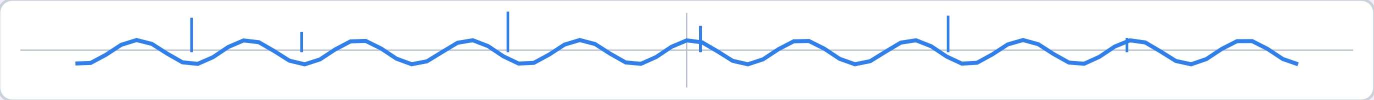
MEANING AND MEMORY

Meaning: A function with at least one point where continuity fails.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
continuity fails at one or more points
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Jump, removable, infinite discontinuity

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Discontinuous without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Removable discontinuity

DIFFICULTY: INTERMEDIATE

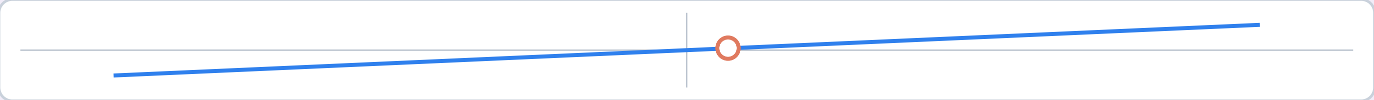
MEANING AND MEMORY

Meaning: A hole that could be fixed by redefining one value.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
limit exists but function value is missing or mismatched
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Limit, hole, redefine value

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Removable discontinuity without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Jump discontinuity

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A break where the left-side and right-side limits do not match.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
left and right limits exist but differ
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Step, regime switch, left/right limits

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Jump discontinuity without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Infinite discontinuity

DIFFICULTY: INTERMEDIATE

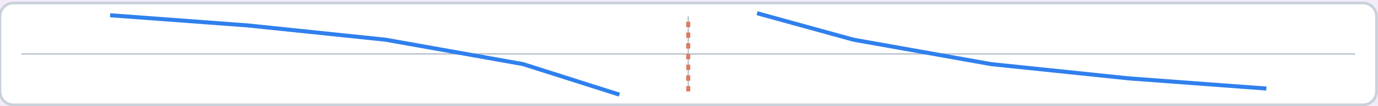
MEANING AND MEMORY

Meaning: A break where the function blows up without bound near a point.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
function grows without bound near a point
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Asymptote, reciprocal, blow-up

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Infinite discontinuity without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Smooth

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function with derivatives available to the needed order.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
has derivatives of all required orders
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Differentiable, optimization, splines

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Smooth without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Differentiable

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A function with a well-defined local slope.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
local derivative f'(x) exists
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: Derivative, gradient, smoothness

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Differentiable without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Non-differentiable

DIFFICULTY: INTERMEDIATE

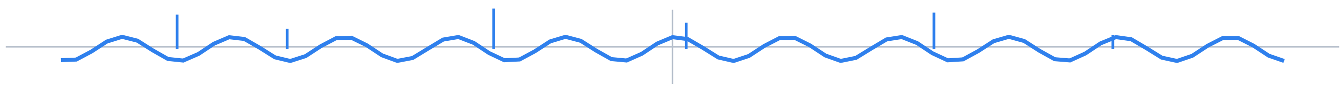
MEANING AND MEMORY

Meaning: A function where local slope fails because of a kink, jump, cusp, or asymptote.
Memory jogger: Think: what happens under tiny input changes.

FORMULA AND BEHAVIOR

```
derivative fails at a kink, cusp, jump, or vertical tangent
```

Formula intuition: Optimization, interpolation, and simulation stability depend heavily on how the function behaves near boundaries and tiny perturbations.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to decide whether an objective, rule, or feature is stable under small input changes and friendly to gradient-based methods.
Simple market example: Example: a hard threshold can cause unstable turnover when the score jitters around the cutoff.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Write a pure function, vectorize it for arrays, validate domain assumptions, and add shape tests with synthetic inputs before using live data.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: A graph that looks smooth at screen resolution can still have mathematical or numerical discontinuities.
Related terms: ReLU, absolute value, kink

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Non-differentiable without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 09

Composition, Representations, Sequences, and Series

These are the construction tools for chaining transforms, describing trajectories, and turning time-indexed updates into research logic.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Composite

DIFFICULTY: INTERMEDIATE

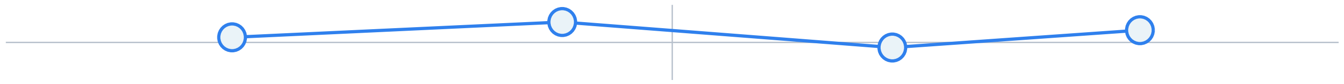
MEANING AND MEMORY

Meaning: A function made by feeding the output of one function into another.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
(f o g)(x)=f(g(x))
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Pipeline, transformation chain, inverse

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Composite without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Inverse

DIFFICULTY: INTERMEDIATE

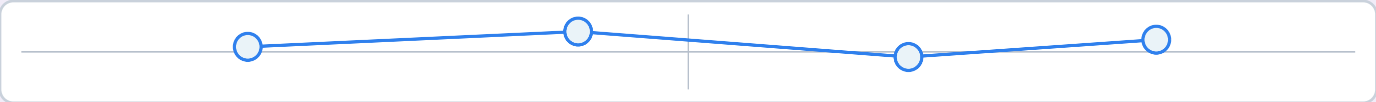
MEANING AND MEMORY

Meaning: A function that reverses another function on an appropriate domain.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
f^-1(y)=x when f(x)=y
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Invertible, bijective, solve-for-input

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Inverse without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Implicit

DIFFICULTY: INTERMEDIATE

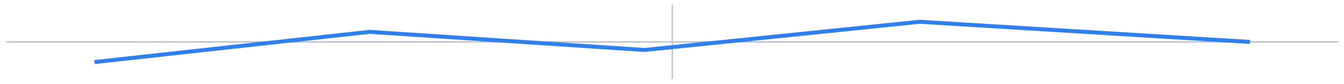
MEANING AND MEMORY

Meaning: A relation where output is defined indirectly by an equation.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
F(x,y)=0 defines y indirectly
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Constraint, root finding, level set

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Implicit without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Explicit

DIFFICULTY: INTERMEDIATE

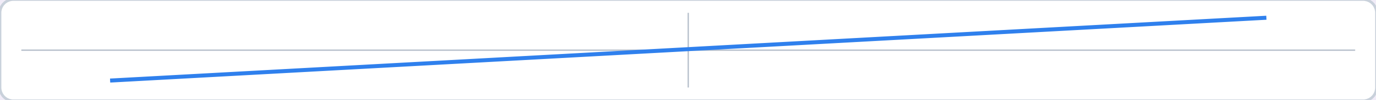
MEANING AND MEMORY

Meaning: A rule that directly gives output as a function of input.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
y=f(x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Direct rule, prediction function

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Explicit without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

COMPOSITION, REPRESENTATIONS, SEQUENCES, AND SERIES

Parametric

DIFFICULTY: INTERMEDIATE

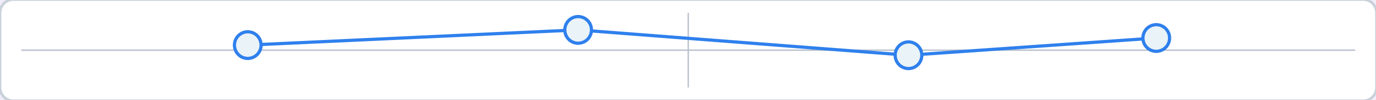
MEANING AND MEMORY

Meaning: A representation where coordinates are functions of a separate parameter.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
x=x(t), y=y(t)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Trajectory, curve, simulation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Parametric without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Polar

DIFFICULTY: INTERMEDIATE

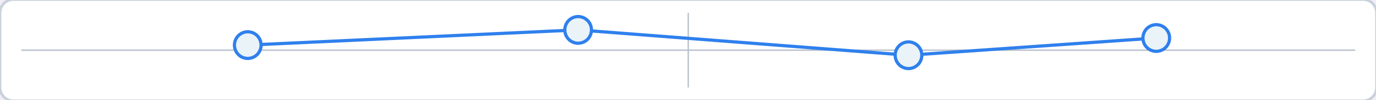
MEANING AND MEMORY

Meaning: A representation where distance from origin depends on angle.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
r=f(theta)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Angle, radius, cyclic data

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Polar without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Recursive

DIFFICULTY: INTERMEDIATE

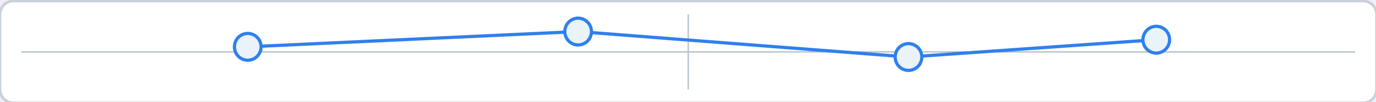
MEANING AND MEMORY

Meaning: A rule that defines each value from earlier values.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
a_n = F(a_{n-1}, a_{n-2}, ...)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sequence, state update, AR process

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Recursive without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sequence

DIFFICULTY: INTERMEDIATE

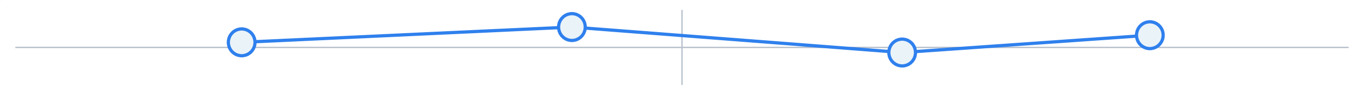
MEANING AND MEMORY

Meaning: A function whose domain is usually the positive integers.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
ordered values a_1, a_2, ...
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Time index, recursive, series

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sequence without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Series

DIFFICULTY: INTERMEDIATE

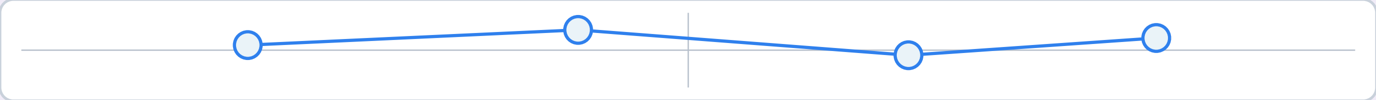
MEANING AND MEMORY

Meaning: A sum of sequence terms.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
sum of sequence terms
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Sum, convergence, cumulative return

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Series without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Arithmetic sequence

DIFFICULTY: INTERMEDIATE

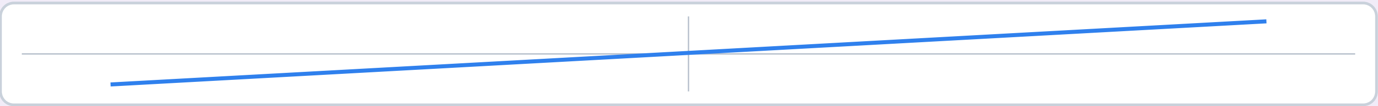
MEANING AND MEMORY

Meaning: A sequence that changes by a constant difference.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
a_n=a_1+(n-1)*d
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Linear trend, constant increment

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Arithmetic sequence without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Geometric sequence

DIFFICULTY: INTERMEDIATE

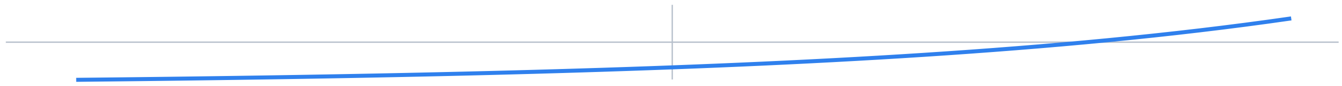
MEANING AND MEMORY

Meaning: A sequence that changes by a constant ratio.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
a_n=a_1*r^(n-1)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Compounding, exponential growth

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Geometric sequence without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Fibonacci sequence

DIFFICULTY: INTERMEDIATE

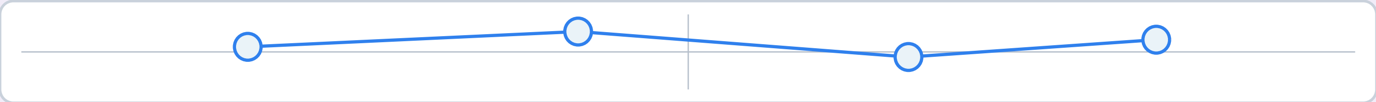
MEANING AND MEMORY

Meaning: A recursive sequence where each term is the sum of the previous two.
Memory jogger: Think: how functions are built, chained, or indexed.

FORMULA AND BEHAVIOR

```
F_n=F_{n-1}+F_{n-2}
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to build pipelines: raw prices to returns, returns to features, features to scores, scores to signals, signals to portfolio actions.
Simple market example: Example: price -> return -> rolling z-score -> sigmoid -> position is a composed research pipeline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Represent the function as a node in a directed pipeline so data lineage, parameter values, and transformations are auditable.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Recursion, ratio, toy model

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Fibonacci sequence without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 10

Probability and Distribution Functions

Distribution functions connect mathematical shape to uncertainty, likelihood, simulation, calibration, and risk measurement.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Probability density

DIFFICULTY: INTERMEDIATE

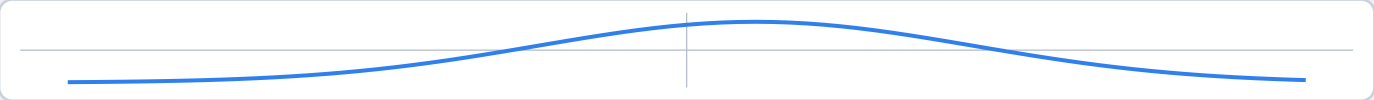
MEANING AND MEMORY

Meaning: A nonnegative function whose area over an interval gives probability.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
P(a<=X<=b)=integral_a^b f(x) dx
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: simulate returns from a distribution and compare realized drawdowns against the simulated risk envelope.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: CDF, likelihood, distribution

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Probability density without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Cumulative distribution

DIFFICULTY: INTERMEDIATE

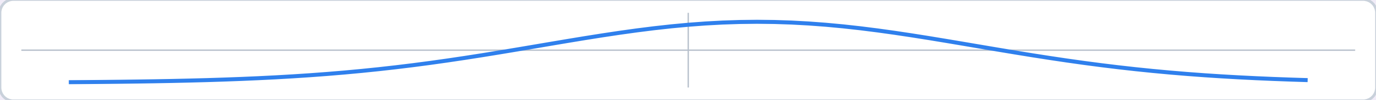
MEANING AND MEMORY

Meaning: A function returning the probability that a random variable is at or below x.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
F(x)=P(X<=x)
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: simulate returns from a distribution and compare realized drawdowns against the simulated risk envelope.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: PDF, quantile, probability integral transform

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Cumulative distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Normal distribution

DIFFICULTY: INTERMEDIATE

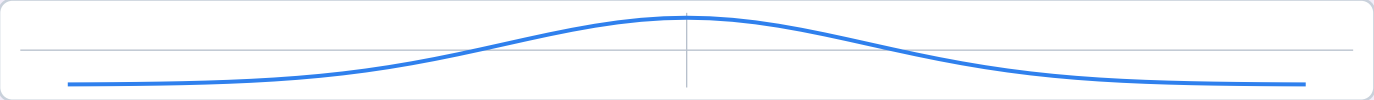
MEANING AND MEMORY

Meaning: A bell-shaped probability distribution controlled by mean and standard deviation.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
f(x)=1/(sigma*sqrt(2*pi))*e^(-(x-mu)^2/(2*sigma^2))
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: z = (return - mean)/std compares an observed return to a bell-curve baseline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Normality is a model assumption; markets often show fat tails, skew, and regime changes.
Related terms: Gaussian, error function, z-score

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Normal distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Uniform distribution

DIFFICULTY: INTERMEDIATE

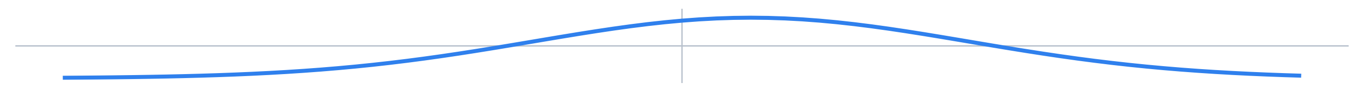
MEANING AND MEMORY

Meaning: A distribution where all values in an interval are equally likely.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
f(x)=1/(b-a) for a<=x<=b
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: simulate returns from a distribution and compare realized drawdowns against the simulated risk envelope.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Simulation, random baseline, bounded support

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Uniform distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Binomial distribution

DIFFICULTY: INTERMEDIATE

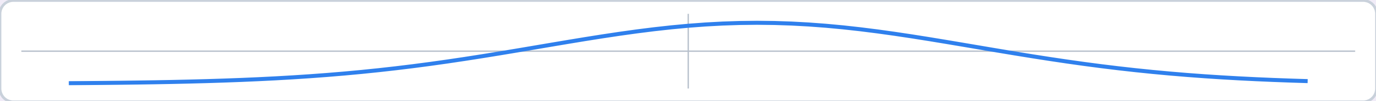
MEANING AND MEMORY

Meaning: A count distribution for successes in a fixed number of independent trials.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
P(X=k)=C(n,k)*p^k*(1-p)^(n-k)
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: compare observed win count against an assumed hit rate over n trades.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Do not memorize the shape only. Track domain, range, parameters, and what the transform does to noise.
Related terms: Bernoulli, hit rate, trials

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Binomial distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

PROBABILITY AND DISTRIBUTION FUNCTIONS

Poisson distribution

DIFFICULTY: INTERMEDIATE

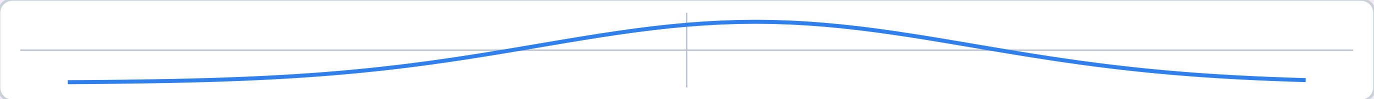
MEANING AND MEMORY

Meaning: A count distribution for event arrivals over a fixed exposure window.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
P(X=k)=e^-lambda*lambda^k/k!
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: model the count of news events, quote updates, or fills arriving in a fixed interval.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Poisson assumes a stable intensity over the window; clustered events break that assumption.
Related terms: Arrival count, intensity, rare events

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Poisson distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Exponential distribution

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A waiting-time distribution with memoryless decay.
Memory jogger: Think: shape of uncertainty.

FORMULA AND BEHAVIOR

```
f(x)=lambda*e^(-lambda*x), x>=0
```

Formula intuition: A distribution function must be tied to support, parameters, assumptions, and estimation method; otherwise the curve is just decoration.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it to model uncertainty, calibrate likelihoods, simulate outcomes, evaluate tails, and compare observed data against assumptions.
Simple market example: Example: simulate returns from a distribution and compare realized drawdowns against the simulated risk envelope.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use scipy.stats or tested internal wrappers. Check support, parameterization, random seed, calibration window, and tail behavior.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Memorylessness is a strong assumption; waiting times in markets often cluster.
Related terms: Waiting time, Poisson process, decay

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Exponential distribution without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

SECTION 11

Special Functions, Approximation, and ML Activations

Special functions and activations appear in numerical libraries, probability models, kernels, optimizers, and nonlinear machine-learning systems.

Concepts in this section

Read the cards vertically. Use the quick recall line as a self-test before moving to the next concept.

Gamma

DIFFICULTY: INTERMEDIATE

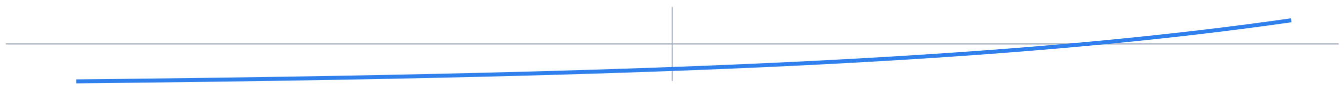
MEANING AND MEMORY

Meaning: A continuous extension of factorial and a core special function in probability.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
Gamma(z)=integral_0^inf t^(z-1)*e^-t dt
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Gamma has multiple parameterizations in software; check shape-rate versus shape-scale.
Related terms: Factorial, gamma distribution, beta

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Gamma without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Beta

DIFFICULTY: INTERMEDIATE

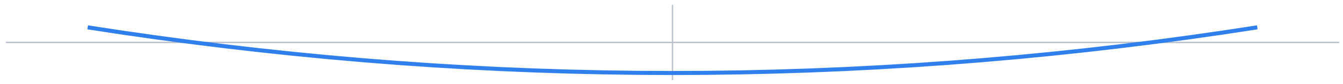
MEANING AND MEMORY

Meaning: A special function that normalizes beta distributions and ratios on bounded intervals.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
B(a,b)=Gamma(a)*Gamma(b)/Gamma(a+b)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Beta is bounded on [0,1]; do not use it directly for unbounded returns.
Related terms: Gamma, beta distribution, bounded uncertainty

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Beta without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Zeta

DIFFICULTY: INTERMEDIATE

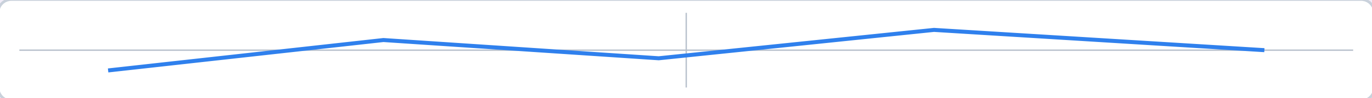
MEANING AND MEMORY

Meaning: A function built from reciprocal powers, central in analytic number theory and some regularization ideas.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
zeta(s)=sum_{n=1}^inf 1/n^s
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Series, convergence, heavy tails

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Zeta without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

DIFFICULTY: INTERMEDIATE

Factorial

MEANING AND MEMORY

Meaning: A product over descending positive integers.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
n! = n*(n-1)*...*1
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Combinatorics, binomial coefficient, gamma

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Factorial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Double factorial

DIFFICULTY: INTERMEDIATE

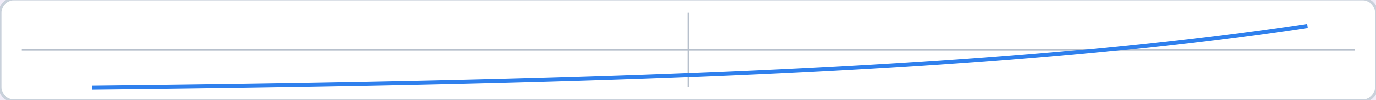
MEANING AND MEMORY

Meaning: A factorial-like product that skips every other integer.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
n!! = n*(n-2)*(n-4)*...
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Combinatorics, normal moments

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Double factorial without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Gamma extension

DIFFICULTY: INTERMEDIATE

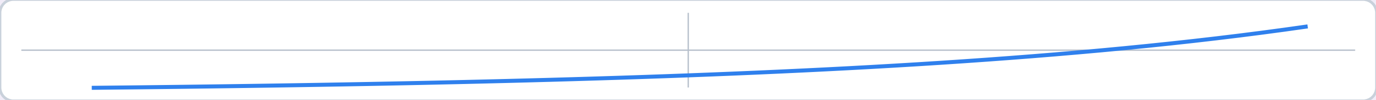
MEANING AND MEMORY

Meaning: The relationship that lets factorial logic extend beyond integers.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
Gamma(n+1)=n!
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Gamma, factorial, continuous interpolation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Gamma extension without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Error function

DIFFICULTY: INTERMEDIATE

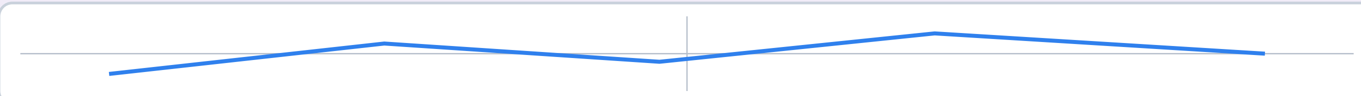
MEANING AND MEMORY

Meaning: An integral of the Gaussian curve used to express normal probabilities.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
erf(x)=2/sqrt(pi)*integral_0^x e^(-t^2) dt
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Normal CDF, Gaussian integral

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Error function without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Bessel

DIFFICULTY: INTERMEDIATE

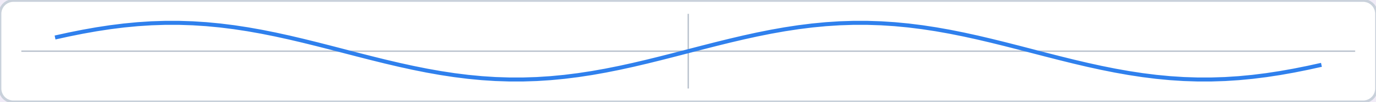
MEANING AND MEMORY

Meaning: A family of oscillatory special functions appearing in radial and cylindrical systems.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
solutions to x^2*y''+x*y'+(x^2-nu^2)*y=0
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Oscillation, radial systems, filters

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Bessel without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Airy

DIFFICULTY: INTERMEDIATE

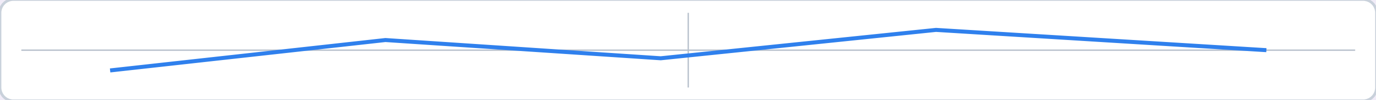
MEANING AND MEMORY

Meaning: A special function family that describes turning-point behavior in differential equations.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
solutions to y'' - x*y = 0
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Differential equations, turning points

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Airy without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Elliptic

DIFFICULTY: INTERMEDIATE

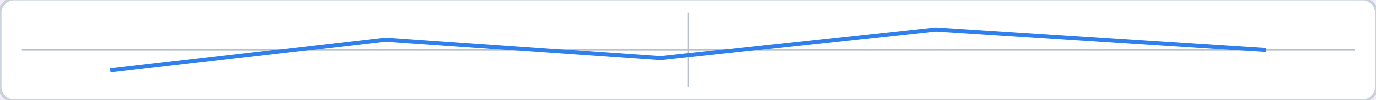
MEANING AND MEMORY

Meaning: A family of functions and integrals extending trigonometric ideas to elliptic geometry.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
integrals that generalize arc length of ellipses
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Special functions, geometry, nonlinear systems

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Elliptic without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Delta

DIFFICULTY: INTERMEDIATE

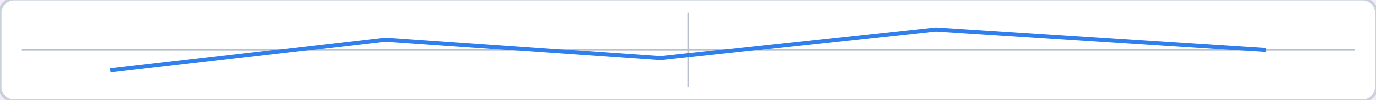
MEANING AND MEMORY

Meaning: An impulse-like generalized function concentrated at one point.
Memory jogger: An idealized point impulse.

FORMULA AND BEHAVIOR

```
Dirac delta: zero except at 0, integral equals 1
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Dirac delta is a generalized function, not an ordinary finite-height spike.
Related terms: Impulse, convolution, point mass

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Delta without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Sinc

DIFFICULTY: INTERMEDIATE

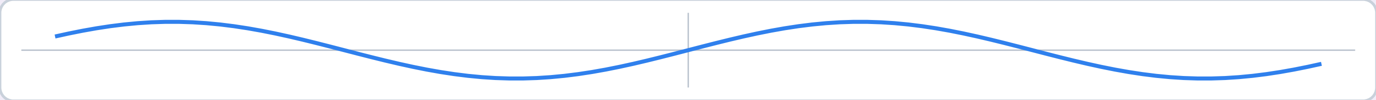
MEANING AND MEMORY

Meaning: A normalized sine-over-input function central to sampling and frequency response.
Memory jogger: The sampling ripple.

FORMULA AND BEHAVIOR

```
sinc(x)=sin(pi*x)/(pi*x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Sampling, Fourier transform, interpolation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Sinc without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Gaussian

DIFFICULTY: INTERMEDIATE

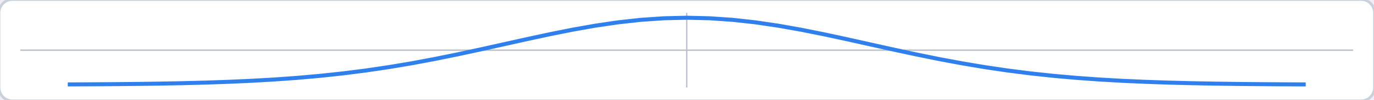
MEANING AND MEMORY

Meaning: A bell-shaped exponential function used as a density, kernel, and smoothing curve.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
G(x)=A*e^(-(x-mu)^2/(2*sigma^2))
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: z = (return - mean)/std compares an observed return to a bell-curve baseline.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Normal distribution, kernel, smoothing

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Gaussian without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Weierstrass

DIFFICULTY: INTERMEDIATE

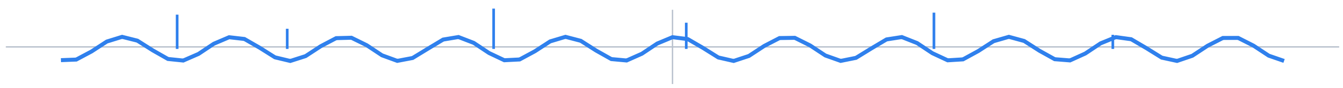
MEANING AND MEMORY

Meaning: A classic example of a function that is continuous everywhere but differentiable nowhere.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
sum a^n*cos(b^n*pi*x), continuous but nowhere differentiable
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Continuity, non-differentiability, fractal behavior

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Weierstrass without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Lambert W

DIFFICULTY: INTERMEDIATE

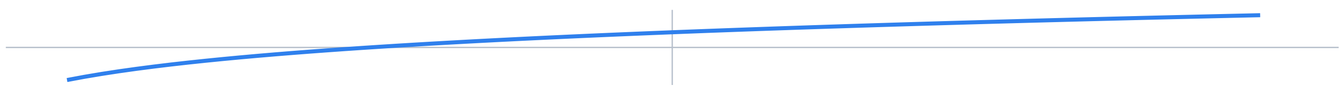
MEANING AND MEMORY

Meaning: An inverse function for expressions where a variable appears both outside and inside an exponential.
Memory jogger: Undo x*exp(x).

FORMULA AND BEHAVIOR

```
W(z)*e^W(z)=z
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Inverse exponential, solve nonlinear equations

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Lambert W without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Floor-step hybrid

DIFFICULTY: INTERMEDIATE

MEANING AND MEMORY

Meaning: A custom bucketed rule that combines integer bins and threshold switches.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
combine floor buckets with step thresholds
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.

QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Floor, step, bucketed rule

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Floor-step hybrid without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Spline

DIFFICULTY: INTERMEDIATE

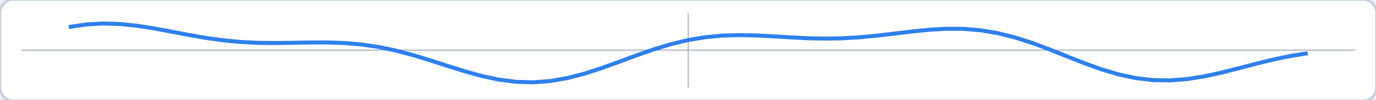
MEANING AND MEMORY

Meaning: A smooth curve assembled from polynomial pieces.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
piecewise polynomials joined at knots
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Piecewise, smooth, interpolation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Spline without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Bezier

DIFFICULTY: INTERMEDIATE

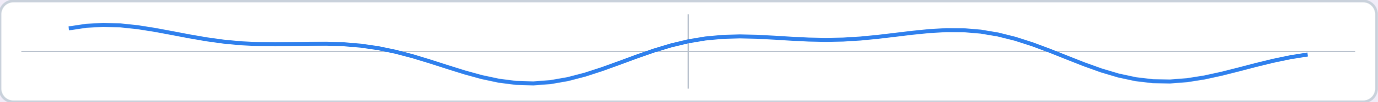
MEANING AND MEMORY

Meaning: A parametric curve controlled by anchor points.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
B(t)=sum P_i*Bernstein_i,n(t)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Parametric, spline, curve design

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Bezier without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Activation function

DIFFICULTY: INTERMEDIATE

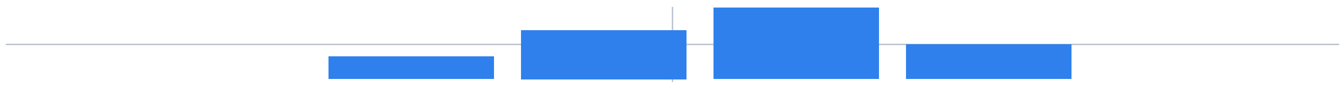
MEANING AND MEMORY

Meaning: A nonlinear transform that gives a model expressive power.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
nonlinear transform inside a neural network
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: ReLU, sigmoid, tanh, softmax

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Activation function without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Softmax

DIFFICULTY: INTERMEDIATE

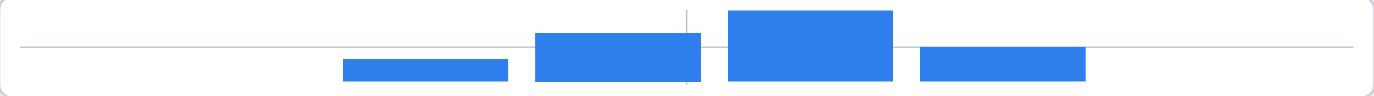
MEANING AND MEMORY

Meaning: A vector function that converts scores into normalized probabilities.
Memory jogger: Turn a score vector into shares of one total.

FORMULA AND BEHAVIOR

```
softmax(z_i)=e^(z_i)/sum_j e^(z_j)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: convert model scores for long, flat, and short into allocation probabilities that sum to one.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Use numerically stable forms: subtract max before softmax and avoid overflow in exp for large positive or negative scores.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Softmax probabilities are relative to the scores in the same vector, not absolute truth probabilities.
Related terms: Activation, probability vector, classifier

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Softmax without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

ReLU

DIFFICULTY: INTERMEDIATE

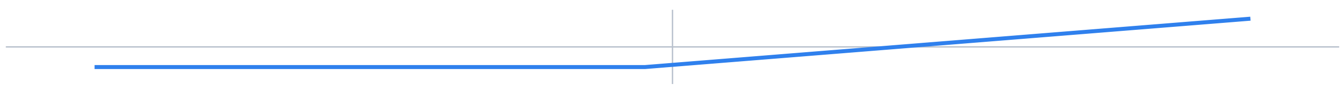
MEANING AND MEMORY

Meaning: A piecewise linear activation that zeros negative values and passes positives through.
Memory jogger: Negative off, positive on.

FORMULA AND BEHAVIOR

```
ReLU(x)=max(0,x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: a neural feature is inactive for negative evidence and grows linearly only when evidence turns positive.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: ReLU is not differentiable at zero, even though most frameworks use a convention for gradients.
Related terms: Ramp, activation, non-differentiable at zero

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of ReLU without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Leaky ReLU

DIFFICULTY: INTERMEDIATE

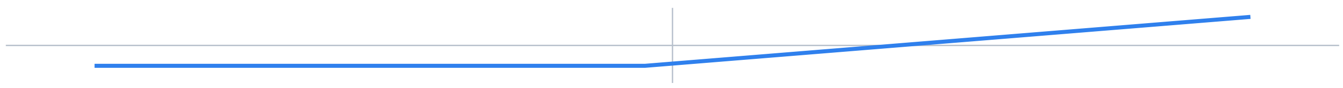
MEANING AND MEMORY

Meaning: A ReLU variant that leaves a small slope for negative inputs.
Memory jogger: Think: specialized math object used by numerical systems.

FORMULA AND BEHAVIOR

```
max(alpha*x, x)
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: The negative slope alpha is a model choice, not a universal constant.
Related terms: ReLU, activation, dying neuron fix

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Leaky ReLU without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Tanh activation

DIFFICULTY: INTERMEDIATE

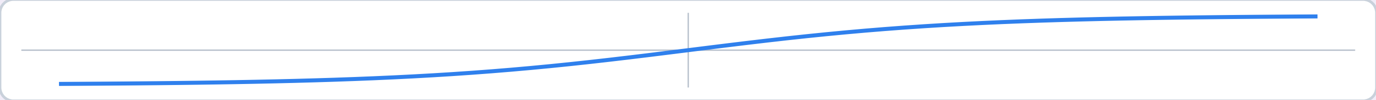
MEANING AND MEMORY

Meaning: A tanh nonlinearity used to create bounded positive and negative model outputs.
Memory jogger: Bounded signed signal.

FORMULA AND BEHAVIOR

```
tanh(x), usually scaled to [-1,1]
```

Formula intuition: Track domain, range, numerical stability, and parameter conventions before wiring the function into a research pipeline.



QUANT USE AND SIMPLE EXAMPLE

Quant use: Use it when standard elementary functions are not enough: numerical modeling, probabilistic normalization, kernels, activations, and approximation systems.
Simple market example: Example: use the library implementation directly, then validate domains, overflow behavior, and numerical stability before trusting results.

SYSTEM FIT AND IMPLEMENTATION

Research-system fit: Store it as a reusable transform object with metadata: domain, range, parameters, differentiability, monotonicity, failure modes, and test cases.
Python/system implementation idea: Prefer scipy.special, numpy, or framework-native kernels. Add domain tests and compare against known values before production use.

TRAPS, RELATED TERMS, AND EDGE CASES

Confusions to avoid: Library names, scaling conventions, and parameter order can differ; verify with known values.
Related terms: Hyperbolic tangent, bounded signal, activation

REVIEW PROMPT

Quick recall: Can you name the domain, range, main formula, and one quant use of Tanh activation without looking? Then test it on x = -2, 0, 0.5, 1, and 2 where valid.

Final review checklist

Use this to decide which function family fits a research problem.

NEED A BASIC CURVE OR APPROXIMATION?

Start with linear, polynomial, power, root, reciprocal, rational, or spline functions.

NEED A RULE, BUCKET, OR ON/OFF DECISION?

Use piecewise, threshold, step, indicator, floor, ceiling, clamp, or modulo logic.

NEED COMPOUNDING, DECAY, OR SCALE COMPRESSION?

Use exponential, natural exponential, exponential decay, logarithmic, natural logarithm, logistic, sigmoid, or softplus.

NEED SEASONALITY OR REPEATED STRUCTURE?

Use sine, cosine, periodic, sinusoidal, square wave, triangle wave, sawtooth wave, or cyclic modulo features.

NEED TO KNOW WHETHER INFORMATION IS PRESERVED?

Check injective, surjective, bijective, one-to-one, many-to-one, invertible, monotonic, and domain restrictions.

NEED OPTIMIZER-FRIENDLY BEHAVIOR?

Prefer continuous, smooth, differentiable functions; isolate discontinuities and non-differentiable kinks.

NEED UNCERTAINTY MODELING?

Use PDFs, CDFs, and named distributions, but validate support, tails, independence assumptions, and parameter stability.

NEED NONLINEAR ML BEHAVIOR?

Use activations such as sigmoid, tanh, ReLU, leaky ReLU, softplus, and softmax with numerical-stability checks.

Common failure modes

The function is rarely the whole problem. The implementation context matters.

INVALID DOMAIN

Log of nonpositive values, square root of negatives, inverse trig outside support, acosh below 1, and atanh outside (-1,1).

BOUNDARY INSTABILITY

Hard thresholds, clamps, and step functions can flip repeatedly when noisy values hover near a cutoff.

OVERFLOW AND UNDERFLOW

Exp, sigmoid, softmax, gamma, factorial, and zeta-style computations can exceed floating-point limits.

FALSE SMOOTHNESS

A plotted curve can look smooth while the formula has kinks, jumps, holes, or pathological behavior.

BACKTEST LEAKAGE

Any function with fitted parameters must be fit on training data only, then applied out-of-sample.

CONVENTION MISMATCH

Software may differ in log base, branch cuts, inverse-function range, rounding negative numbers, or distribution parameterization.

OVERFITTED FLEXIBILITY

High-degree polynomials, splines with too many knots, and complex activations can memorize noise.

UNTRACKED TRANSFORM LINEAGE

If a function is not versioned with parameters and code, the signal may not be reproducible.

Concept index

Every requested function term is included with its card page number.

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